

Thanh Nam Tran

Silver Spring, MD 20904

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Summary

A generalist and quick learner with full-stack development experience building ERP applications, APIs, data dashboards, and cloud deployments. Skilled across frontend, backend, databases, and DevOps, with additional experience in AI/ML research, model evaluation, and intelligent application development.

Technical Skills

Programming Languages: Python, JavaScript, C++

Frontend: React, Next.js, HTML, CSS, Tailwind CSS, Bootstrap

Backend & APIs: Node.js, Express.js, Django, Flask, FastAPI, Odoo ERP, REST APIs, WebSockets

Databases: PostgreSQL, Microsoft SQL Server, MongoDB, Supabase, Django ORM

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Git, GitHub, Vercel

AI/ML & Data: Pytorch, scikit-learn, OpenCV, SpaCy, LangChain, Retrieval-Augmented Generation (RAG), Vector Embeddings, Fine-Tuning, Model Evaluation, Scikit-learn, Pandas, NumPy, Power BI, Tableau

Experience

Software Developer | Media Cybernetics, Inc | Rockville, MD – 02/2022– Present

- Build and maintain full-stack ERP applications in Odoo, developing frontend interfaces, backend business logic, database workflows, system integrations, and reusable modules.
- Support reliable deployments through CI/CD, Docker, Kubernetes, and performance monitoring
- Developed a company metrics dashboard using REST APIs, Microsoft SQL Server, Power BI, and batch-processing workflows to automate business reporting.
- Work on the development of an AI-powered chatbot and internal knowledge base using retrieval-augmented generation, vector search, and NLP to improve access to company information and support resources..

Engineering Intern | Media Cybernetics, Inc | Rockville, MD – 05/2021– 12/2021

- Maintain and update legacy code to ensure compatibility between multiple legacy Capture Interface versions.
- Develop and improve Capture Interfaces features using C++, software SDK, and microscope cameras API.

Projects

Share Your Song | Full-Stack Music Community Platform – [GitHub](#) / [Site](#)

Next.js, React, TypeScript, Tailwind CSS, MongoDB, NextAuth, AWS S3, Pusher, Resend

A social music-sharing platform with secure user authentication, customizable profiles, media uploads, song and recommendation posts, comments, real-time chat, activity notifications, and transactional email workflows.

Adapting Vision GNN SAR Iceberg Imagery Classification | Group Research Project – [Report](#) / [Presentation](#)

Vision Graph Neural Networks, Computer Vision, Model Fine-Tuning, Training, and Evaluation

Applied Vision Graph Neural Networks to classify ice, ocean, boundaries, and other regions in Antarctic SAR imagery using patch-level learning, self-supervised training, and graph-based feature analysis.

Education

University of Maryland, Baltimore County (UMBC) 01/2025– Present ▪ M.S. Computer Science - AI/ML Focus

Relevant Courses: Machine Learning, Principles of Artificial Intelligence, Neural Networks, Computer Vision

University of Maryland, Baltimore County (UMBC) 08/2019– 12/2021 ▪ B.S. Computer Science - Data Science Focus

Montgomery College 08/2016– 05/2019 ▪ A.A. Computer Science